B.Tech CSE - Third Year | Lab 3

Assignment 3: Bubble Sort • *Aim: Sort arrays by comparing and swapping adjacent values.*

Experiment No. 3.1: Perform one bubble-sort pass

1. **Aim:** Implement the task specified in the BCS2024545 lab assignment.
2. **Program:**

```
public class Task_01_Perform_one_bubble {
    public static void main(String[] args) {
        int[] a = {5, 1, 4, 2, 8};
        for (int i = 0; i < a.length - 1; i++) {
            if (a[i] > a[i + 1]) {
                int temp = a[i]; a[i] = a[i + 1]; a[i + 1] = temp;
            }
        }
        for (int value : a) System.out.println(value);
    }
}
```

3. **Output:**

```
1
4
2
5
8
```

4. **Viva-Voce:** Why compare neighbours? What is a swap variable used for? Is one pass enough to fully sort every array?

Experiment No. 3.2: Sort an array using bubble sort

1. **Aim:** Implement the task specified in the BCS2024545 lab assignment.
2. **Program:**

```
public class Task_02_Sort_an_array_using_bubble_sort {
    public static void main(String[] args) {
        int[] a = {5, 1, 4, 2, 8};
        for (int pass = 0; pass < a.length - 1; pass++) {
            for (int i = 0; i < a.length - 1 - pass; i++) {
                if (a[i] > a[i + 1]) {
                    int temp = a[i]; a[i] = a[i + 1]; a[i + 1] = temp;
                }
            }
        }
        for (int value : a) System.out.println(value);
    }
}
```

3. **Output:**

```
1
2
4
5
8
```

4. **Viva-Voce:** Why are two loops used? What does each pass accomplish? What is bubble sort's worst-case complexity?

Experiment No. 3.3: Bubble sort three values

1. **Aim:** Implement the task specified in the BCS2024545 lab assignment.
2. **Program:**

```
public class Task_03_Bubble_sort_three_values {
    public static void main(String[] args) {
        int[] a = {9, 7, 3};
        for (int pass = 0; pass < a.length - 1; pass++) {
            for (int i = 0; i < a.length - 1 - pass; i++) {
                if (a[i] > a[i + 1]) {
                    int temp = a[i]; a[i] = a[i + 1]; a[i + 1] = temp;
                }
            }
        }
        for (int value : a) System.out.println(value);
    }
}
```

3. **Output:**

```
3
7
9
```

4. **Viva-Voce:** Does bubble sort work on descending input? Is it in-place? What is meant by sorted ascending order?